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The Thickness of Mono-age Disk Populations Follows the Curvature of the Total Potential

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ABSTRACT

We test two models that relate the thickness of a mono-age disk population to its vertical velocity dispersion σ_z: an isothermal tracer of the total potential, $\rho \propto \exp(-\Delta\Phi/\sigma_z^2)$, and a self-gravitating sheet, $z_0 = \sigma_z^2/(\pi G\Sigma)$, whose thickness grows as σ_z^2. In seven Auriga disks, the tracer predicts the scale heights of 64 of 70 populations within 10 percent at every age, nearly as well as the Jeans equation, whereas the sheet does so for 8, predicting the youngest populations 0.51 to 0.78 times as thick as measured and the oldest 2.20 to 3.37 times. Although the two models coincide for a self-gravitating layer, each population holds a small part of the mass of its disk and moves in the potential of all of it, so its height grows as σ_z^η, with η of 1.10 to 1.26 set by the curvature of that potential, which makes age-thickness relations grow as age$^{\eta\beta_z}$ rather than age$^{2\beta_z}$ for $\sigma_z \propto$ age$^{\beta_z}$. Their dispersions rise with height. Summed, the tracers reproduce the heights and flares of the whole disks. Near the Sun, Gaia DR3 star counts split by LAMOST and APOGEE ages follow the tracer with no trend with age over 0-10 Gyr, match its level in potentials with the measured local density, and imply midplane densities of 0.089 ± 0.017, 0.104 ± 0.013 and $0.120 \pm 0.030\, M_\odot\,\mathrm{pc}^{-3}$ from LAMOST and two APOGEE age scales, consistent with the measured 0.097 ± 0.013. The heights of Imig et al. (2025) agree with our counts in level but grow more slowly with age.

Keywords: Galaxy dynamics (591) — Galaxy structure (622) — Milky Way dynamics (1051) — Stellar dynamics (1596)

1. INTRODUCTION

The vertical structure of a population of stars of one age records how the disk heated it. Both its dispersion (J. Holmberg et al. 2009; Y.-S. Ting & H.-W. Rix 2019; S. Sharma et al. 2021) and its thickness grow with age, the thickness now measured population by population, split by abundance or by age, in large spectroscopic and astrometric surveys (J. Bovy et al. 2012b, 2016; J. T. Mackereth et al. 2017; D. Mirabal et al. 2026). Turning one into the other needs a model of the vertical structure, of which two are in use. In the first, a population of one age is an isothermal tracer of the total gravitational potential, whose density falls as $\exp[-\Delta\Phi(R,z)/\sigma_z^2]$ with the vertical potential $\Delta\Phi$ of all the mass. Models of the Milky Way disk have long been built this way, from a sum of such populations with dispersions prescribed by an age-velocity relation and compared with star counts that mix all ages (O. Bienaymé et al. 1987; A. C. Robin et al. 2003; A. Just & H. Jahreiß 2010; K. Sysoliatina & A. Just 2021, 2022). The density of a single tracer has been predicted from its kinematics in this way (J. N. Bahcall 1984; J. Holmberg & C. Flynn 2000). The inverse problem, inferring the potential near the Sun from the density and the kinematics of a tracer, goes back to J. C. Kapteyn (1922) and J. H. Oort (1932, 1960) and was taken up with K dwarfs toward the south Galactic pole by K. Kuijken & G. Gilmore (1989a,b). Mono-abundance populations, whose vertical dispersions change little with height (J. Bovy et al. 2012a), have been used in the same direction (J. Bovy & H.-W. Rix 2013).

In the second model, a population is a self-gravitating isothermal sheet of surface density Σ, with the profile $\mathrm{sech}^2(z/z_0)$ and $z_0 = \sigma_z^2/(\pi G\Sigma)$ (L. Spitzer 1942; P. C. van der Kruit & L. Searle 1981). The sheet is the tracer of a layer whose own mass makes its potential, which is how it describes the old disk as a whole. Studies of disk heating and radial migration have adopted it for single populations as well, mostly as a scaling between dispersion and thickness (R. Roškar et al. 2013; M. Martig et al. 2014; R. J. J. Grand et al. 2016; J. T. Mackereth et al. 2019). M. Martig et al. (2026, Sect. 5.3), for example, use it to turn a dispersion that grows as age$^{\beta_z}$ into an expected thickness that grows as age$^{2\beta_z}$. The two models agree for a population that makes its own po-

tential, and differ for one that holds a small part of the mass of its disk. At fixed Σ the sheet makes every population grow as σ_z^2, whereas a tracer grows as σ_z^η, with η set by the shape of $\Delta\Phi$ over the heights that the population reaches: $\eta = 1$ where the potential is harmonic and $\eta = 2$ where it grows linearly with height (P. C. van der Kruit 1988). A. D. Ludlow et al. (2021) interpolate between these limits to predict the half-mass heights of idealized single-component disks from their dispersions. A. Benítez-Llambay et al. (2018) find the same transition in the thickness of gaseous disks. To our knowledge, neither model has been tested quantitatively, population by population, against the heights of mono-age populations measured independently of it. The models of the Milky Way are fitted to counts that mix the populations, with dispersions from an age-velocity relation, whereas the inverse problem takes the heights as data rather than predicting them.

The question also bears on the relation between radial action and scale height, which Y. Jia et al. (2023) find in the Milky Way, Y. Jia et al. (2024) recover in an N-body disk, and P. A. Palicio et al. (2025) find across 23 simulated disks, seven of them from Auriga. Y. Jia et al. (2023) read the relation through the radial action of an epicycle and the thickness of a layer at fixed midplane density. Its slope has been derived for a whole disk from the radial action of an epicycle and the scale height of a sheet (C. Vera-Ciro, in press), so that the age dependence of the slope for a single population depends on which vertical model holds.

Here we test the tracer and the sheet population by population. In seven disks of the Auriga simulations (R. J. J. Grand et al. 2017, 2024), whose heights we measure from the star particles and whose potentials we fit to all the particles, we predict the scale height of every population of 1 Gyr of ages from its own dispersion, and compare the predictions with the measured heights, with the vertical Jeans equation, which assumes nothing about the kinematics, as the reference for any such test. We measure how the heights grow with the dispersion, and compare that growth with the local slope of the tracer of each potential. In the solar neighborhood, we split star counts of Gaia giants into populations with the ages of LAMOST and APOGEE, and test the tracer in a family of potentials that match the measured density near the Sun. We describe the vertical structure of a tracer and our estimator in Section 2, the Auriga populations in Section 3, whole disks in Section 4, the Milky Way in Sections 5 and 6, and discuss the consequences in Section 7.

2. THE VERTICAL STRUCTURE OF A TRACER

2.1. *The tracer and the sheet*

A population whose distribution function depends on the vertical energy as $\exp(-E_z/\sigma_z^2)$ has, in the vertical potential $\Delta\Phi(z) = \Phi(R,z) - \Phi(R,0)$ at a radius R, the density

$$\rho(z) \propto \exp\left[-\frac{\Delta\Phi(z)}{\sigma_z^2}\right], \tag{1}$$

with the same dispersion σ_z at every height. We call this population an isothermal tracer, and take $\Delta\Phi$ from the potential of all the mass. When the population makes the potential itself, the Poisson equation closes Eq. (1) into the sheet of L. Spitzer (1942),

$$\rho(z) \propto \operatorname{sech}^2\left(\frac{z}{z_0}\right), \qquad z_0 = \frac{\sigma_z^2}{\pi G \Sigma}. \tag{2}$$

Above z_0 the sheet falls as an exponential of scale $z_0/2$, which is why the relation is also written $\sigma_z^2 = 2\pi G\Sigma h$ for the scale h of that tail. Because the two scales differ by a factor of 2, we take neither by convention: our estimator (Section 2.5) returns z_0 within 0.3 percent when applied to a sheet, so a population that is a sheet would have $h_Z = z_0$, which is the height the sheet predicts. Applied to a population of one age, Σ is the surface density of all the mass near the plane, which we take within $|z| < 1.1\,\mathrm{kpc}$, so that the sheet predicts $h_Z \propto \sigma_z^2$ for every population of a disk.

2.2. *The vertical Jeans equation*

The tracer solves the vertical Jeans equation of a steady axisymmetric population of density ν,

$$\frac{\partial(\nu\sigma_z^2)}{\partial z} + \frac{1}{R}\frac{\partial(R\nu\sigma_{Rz}^2)}{\partial R} = -\nu\frac{\partial\Phi}{\partial z}, \tag{3}$$

under two assumptions: that σ_z does not change with height, and that the second term, the tilt term, vanishes, where $\sigma_{Rz}^2 \equiv \langle v_R v_z\rangle$ is the covariance of the radial and vertical velocities. Writing the tilt term as $\nu t(z)$, with $t(z) = (R\nu)^{-1}\,\partial(R\nu\sigma_{Rz}^2)/\partial R$ an acceleration that adds to the vertical force, Eq. (3) integrates to

$$\nu(z) = \frac{\sigma_z^2(0)}{\sigma_z^2(z)}\exp\left[-\int_0^z \frac{\partial\Phi/\partial z' + t(z')}{\sigma_z^2(z')}\,dz'\right], \tag{4}$$

which is Eq. (1) when σ_z is constant and $t = 0$. With $\sigma_z(z)$ and $t(z)$ measured at every height, Eq. (4) assumes nothing about the distribution function beyond a steady state and axisymmetry, so that its scatter about the measured heights is the reference for any prediction from the kinematics, although it also carries the noise of the dispersion and the tilt term measured in every slice. In N-body disks, a dispersion that changes with height

improves estimates of the local density from one tracer (S. Garbari et al. 2011), as does the tilt term (S. Sivertsson et al. 2022). The tracer (Eq. 1), the sheet (Eq. 2) and the Jeans equation (Eq. 4) are the three predictions that we set against the measured heights.

2.3. *The local slope of the heights*

For a tracer in a potential $\Delta\Phi \propto |z|^p$, Eq. (1) gives a height that grows as $\sigma_z^{2/p}$. We define the local slope

$$\eta = \frac{d\ln h_Z}{d\ln \sigma_z}, \tag{5}$$

which is 1 in a harmonic potential, as near the midplane of any smooth distribution of mass, and 2 in a potential that grows linearly with height, as above a thin one (P. C. van der Kruit 1988). A real disk has a harmonic core and a linear rise, so η grows with σ_z from 1 toward 2 as the populations reach heights where the potential is no longer harmonic. If $\sigma_z \propto \mathrm{age}^{\beta_z}$, Eq. (5) gives $d\ln h_Z/d\ln \mathrm{age} = \eta\beta_z$, so that a tracer thickens as $\mathrm{age}^{\eta\beta_z}$ over a range of σ_z in which η is constant, whereas the sheet at fixed Σ gives $\mathrm{age}^{2\beta_z}$. At heights comparable to the radius, where the vertical force falls with height, $\Delta\Phi$ grows more slowly than $|z|$, so η would exceed 2 there. In the fitted potentials of the Auriga disks, the local slope of the tracer is at most 1.38 to 1.59 at the dispersions from 5 to 150 $\mathrm{km\,s^{-1}}$ at which we measure it (Section 3.3).

2.4. *What a test of the tracer can show*

Part of any agreement between the tracer and the measured heights is guaranteed. In a harmonic potential $\Delta\Phi = \Omega_z^2 z^2/2$, with vertical frequency Ω_z, the vertical virial theorem of a steady population gives $\langle v_z^2\rangle = \Omega_z^2\langle z^2\rangle$ whatever its distribution function, so the rms height of a cold population whose dispersion near the plane represents its stars follows from its vertical velocities as Eq. (1) gives it. Its windowed scale height also depends on the shape of its profile: populations whose distribution functions of the vertical energy are $\exp[-(E_z/E_0)^q]$ with q of 0.7 and 1.5 have 0.85 and 1.14 times the height of the isothermal one with the same $\langle v_z^2\rangle$, at every dispersion. The hot populations, which reach heights where the potential is no longer harmonic, depend on both the shape and the potential. We therefore report the cold and the hot populations separately.

2.5. *The scale height and its window*

We measure the scale height of a set of heights $|z|$, or of a profile, by maximum likelihood of an exponential truncated at a window L. Its mean height inside the window is

$$\langle z\rangle = h - \frac{L}{e^{L/h}-1}, \tag{6}$$

which we solve for h. The maximum-likelihood estimate of the rate $1/h$ is positive only when $\langle z\rangle < L/2$ (W. L. Deemer & D. F. Votaw 1955, Sect. 3), so a window whose mean reaches $L/2$ gives no height. As P. A. Palicio et al. (2025) do, we average the estimates within L and within $L/2$. A profile that is not exponential has no single scale, so its fitted height depends on the window: a window narrower than the profile reads its flat core as a large scale (Appendix A). We therefore scale the window with the population, $L = 2h_Z$, taking the smallest L that satisfies this condition, so that populations of one profile shape are fitted over the same part of their profiles. An exponential returns its scale in any such window, whereas the sheet of Eq. (2) returns z_0 within 0.3 percent. A window needs 100 stars inside $L/2$: with fewer, the profile of a hot population is nearly flat inside a small window, so that the noise of the fitted $1/h$ can satisfy the condition by chance far below the true scale (Appendix A). Every prediction but the sheet, whose z_0 the estimator returns within 0.3 percent, passes through the same estimator as the stars. Because no window is free of convention, we also measure the half-mass height, below which half of the mass within $|z| < 12$ kpc lies.

3. MONO-AGE POPULATIONS IN AURIGA

3.1. *Simulations and measurements*

We use the seven halos of the Auriga simulations (R. J. J. Grand et al. 2017) at the standard resolution of the public release (R. J. J. Grand et al. 2024) that P. A. Palicio et al. (2025, Table 1) analyze, Au2, Au3, Au16, Au19, Au23, Au24 and Au27, at redshift zero. For each halo we fit an axisymmetric potential to all the particles of the snapshot with AGAMA (E. Vasiliev 2019), as a CylSpline expansion for the gas and the stars of the disk region and a Multipole expansion for everything else. The star particles of the annulus $7.5 < R < 9$ kpc are split into ten bins of 1 Gyr in age, from 0 to 10 Gyr, which give 70 populations. In each bin, σ_z is the dispersion of v_z of the stars with $|z| < 0.3$ kpc, clipped at four standard deviations estimated from the median absolute deviation, whereas h_Z is the scale height of the heights of all its stars, weighted by their initial mass, in the window of Section 2.5. The tracer of Eq. (1) takes $\Delta\Phi$ from the fitted potential at the median radius of the annulus, whereas the sheet of Eq. (2) takes Σ from all the particles of the annulus within $|z| < 1.1$ kpc. Errors are half the range between the 16th and the 84th percentiles of 300 bootstrap resamplings of the stars of each bin, each of which recomputes every quantity, so that the ratio of a predicted to the measured height carries the correlation of the two. A prediction agrees with a measured

height within x percent when $|\ln h_{\rm pred}/h_Z| \leq x/100$, the band that the figures draw. Every percentage that compares two heights in this work is such a logarithm. The scatter of a set of predictions is the root mean square of $\ln h_{\rm pred}/h_Z$. On a mock snapshot of a known galaxy, the analysis recovers $\Delta\Phi$ within 1.0 percent and the height of every bin within 0.3 percent (Appendix B). In the real snapshots, the potential averaged over azimuth gives tracer heights 0.97 to 1.04 times those of the fit.

3.2. *The tracer, population by population*

Figure 1 sets the three predictions of Section 2, the tracer (Eq. 1), the vertical Jeans equation (Eq. 4) and the sheet (Eq. 2), against the measured heights, one panel each. Every point is the logarithm of a predicted over the measured height of one population, against its age, so that a prediction that matches a population puts it on the line at 0, inside the gray band of agreement within 10 percent. In the top panel, the tracer predicts the height of 64 of the 70 populations within 10 percent, 44 within 5 percent, with ratios of 0.83 to 1.16 and a median of 1.01. The populations thus have the heights that their own dispersions give them in the potential of the whole galaxy, from the youngest to the oldest, whose heights reach 4.6 kpc with σ_z of up to 127 km s^{-1}.

As Section 2 argued, part of this agreement holds for any steady population. Below $h_Z = 1$ kpc, where the potential is close to harmonic, the tracer comes within 10 percent in 30 of 32 populations, with a scatter of 5.1 percent. Above 1.5 kpc it does in 23 of 26, with a scatter that grows to 9.9 percent above 2.5 kpc. Of the populations hotter than 60 km s^{-1} (hollow circles in Figure 1), the tracer predicts 16 of 19 within 10 percent, against 48 of 51 of the cooler ones, with a scatter of 5.3 percent. Section 3.6 repeats the test with the stars on disk orbits alone.

3.3. *The local slope of the heights*

We fit η of Eq. (5) as the slope of the weighted least-squares line of $\ln h_Z$ against $\ln \sigma_z$ across the bins of each halo. Both coordinates carry errors, so each bin is weighted by the inverse of $(\delta h_Z/h_Z)^2 + \eta^2(\delta\sigma_z/\sigma_z)^2$, refitted until η converges. The error is scaled by $\sqrt{\chi^2_\nu}$ where $\chi^2_\nu > 1$. Over 0-10 Gyr, η spans 1.10 to 1.26 across the seven halos (circles in Figure 2), with errors of 0.03 to 0.05, between the value of 1 of a harmonic potential and the value of 2 of a linear one. Because η is a local slope, it changes with the range of ages, to 1.03 to 1.35 over the ranges 0-10, 2-10, 0-8 and 2-8 Gyr.

Figure 2 sets the fitted η against the local slope of the tracer itself (gray lines), which the curvature of each potential fixes. For each halo we compute the tracer of the

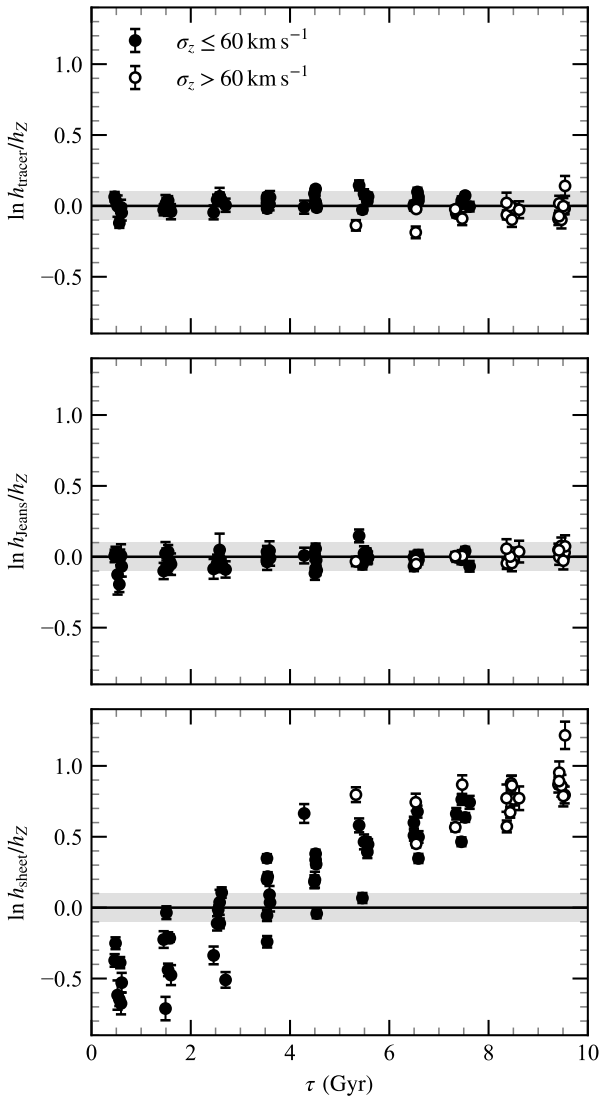


Figure 1. The isothermal tracer predicts the scale height of almost every Auriga population, at every age, about as well as the vertical Jeans equation, whereas the self-gravitating sheet predicts young populations thinner and old ones thicker than measured. Each panel shows, against median age τ, the logarithm of a predicted over the measured scale height h_Z for the 70 populations of 1 Gyr of seven halos: the tracer $\rho \propto \exp(-\Delta\Phi/\sigma_z^2)$ in the potential fitted to all the particles (top), the vertical Jeans equation with the measured $\sigma_z(z)$ and the tilt term, the reference for any prediction from the kinematics (middle), and the sheet $z_0 = \sigma_z^2/(\pi G\Sigma)$ (bottom). Filled circles are the populations with $\sigma_z \leq 60$ km s^{-1}, hollow circles the hotter ones. Errors come from 300 bootstrap resamplings of the stars. The gray band marks agreement within 10 percent, $|\ln h_{\rm pred}/h_Z| \leq 0.1$, which the tracer reaches for 64 populations, the Jeans equation for 65 and the sheet for 8.

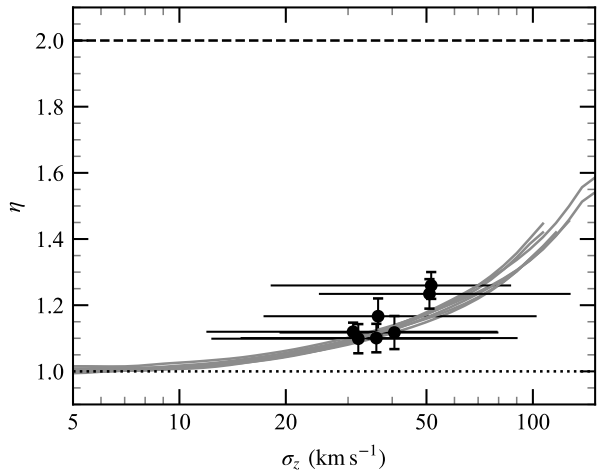


Figure 2. The scale heights grow with the dispersion as $h_Z \propto \sigma_z^\eta$, with the η that the curvature of each potential predicts, rather than the $\eta = 2$ of the sheet. Gray lines show the local slope $\eta = d\ln h_Z / d\ln \sigma_z$ of the tracer in the fitted potential of each halo, which rises from 1 where the potential is harmonic (dotted line) toward 2 where it grows linearly with height (dashed line). Circles show the η fitted to the populations of 0-10 Gyr of each halo, at the geometric mean of their σ_z, with horizontal bars spanning those σ_z. It agrees within two standard errors, in 6 of 7 halos, with the mean slope of its halo's curve over the same dispersions, the slope of the chord between their ends.

fitted potential at 41 dispersions from 5 to 150 km s^{-1}, measure its height in the window, and take the logarithmic derivative. The local slope rises from 1.01 to 1.03 at 10 km s^{-1}, where the populations stay inside the harmonic core of the potential, to 1.11 to 1.13 at 40 km s^{-1}, and to at most 1.38 to 1.59 at the hottest dispersions. Over the dispersions of the populations of each halo, the measured η exceeds the slope of the chord of this curve by -0.03 to 0.11, within two standard errors in 6 of 7 halos. The exception is Au19, whose η of 1.26 ± 0.04 exceeds the chord of 1.15, because its old populations are thicker than the tracer predicts (Section 3.4). The slope stays below 2 over the whole grid: in 5 of 7 halos the window finds no scale from 127 to 150 km s^{-1} on, where $2h_Z$ of the tracer would exceed the 12 kpc of the grid of heights. The hottest population of each of these halos lies a factor of at least 1.29 below the last dispersion at which the window finds a scale. The heights of the populations therefore grow with their dispersions as the curvature of each potential prescribes.

3.4. *Age dependence and the Jeans equation*

The age dependence of the heights tests the tracer more strictly than its scatter does, because a prediction can scatter about the measured heights without a trend, or follow them in scatter but miss how they change with age. We fit the trend p of $h_{\rm pred}/h_Z \propto {\rm age}^p$ by weighted least squares over the bins of each halo, with the error scaled by $\sqrt{\chi^2_\nu}$. A prediction that reproduces how h_Z changes with age has $p = 0$. Over 0-10 Gyr the trend of the tracer lies within 1.7 standard errors of 0 in every halo, with values of -0.045 to 0.014, so that its ratios in the top panel of Figure 1 do not drift with age. Over 2-10 Gyr it is negative in 7 of the 7 halos, a split of signs with a probability of 0.016 under a scatter symmetric about 0. Figure 3 shows these trends, one row per halo (filled circles), next to those of the checks of Sections 3.4 to 3.6, which ask where the residual comes from. The ratio of the tracer to the measured heights therefore declines slightly with age, from a median over the halos of 1.03 at 2-6 Gyr to 0.98 at 8-10 Gyr, by an amount that only some halos resolve on their own.

The vertical Jeans equation tests where that residual comes from. The dispersion rises with height in 70 of 70 populations: at twice the measured height, the dispersion of all the stars of a slice of $|z|$ over that of the slice nearest the plane has a median of 1.27, with a range of 1.02 to 1.73. With the variance of v_z of each population measured in slices of $|z|$ up to 12 kpc, without clipping, the Jeans equation gives heights within 10 percent in 58 populations. Its trends over 2-10 Gyr are negative in 4 of the 7 halos (filled diamonds in Figure 3), with a mean over the halos, weighted by their inverse variances, of -0.013 ± 0.017 against -0.061 ± 0.015 for the tracer, so the pattern of signs of the tracer is gone. In Eq. (4), the prefactor $\sigma_z^2(0)/\sigma_z^2(z)$ lowers the density wherever the dispersion has risen. The rising dispersion therefore makes these heights smaller than those of the tracer in 57 of 70 populations, by a median factor of 0.94 at 2-6 Gyr and 0.99 at 8-10 Gyr, which flattens the decline. The exception is Au19, whose populations older than 5 Gyr reach 0.84 to 0.91 of their measured heights in these Jeans heights. Their tilt term, $(1/R)\,\partial(R\nu\sigma^2_{Rz})/\partial R$, which we fit with straight lines of $\ln\nu$ and σ^2_{Rz} against R over $6 < R < 10.5$ kpc, opposes the vertical force between 1 and 5 kpc, and brings them to 0.96 to 1.08. Their mean vertical motions give thickening rates, $\langle {\rm sign}(z)\, v_z\rangle / \langle |z|\rangle$, of -0.30 to 0.08 per Gyr, with errors of 0.33 to 0.48 per Gyr, within the -1.18 to 1.02 per Gyr of the populations of the same ages in the other halos. This test therefore finds no departure from a steady state in Au19, although its errors allow the mean height to change by tens of percent per Gyr. With the tilt term, the Jeans heights lie within 10 percent in 65 populations, with ratios of 0.82 to 1.16 and a scatter of 5.5 percent (Figure 1, middle). Their trends over 2-10 Gyr, however, are positive in 7 of the 7 ha-

los (hollow diamonds), each within 1.5 standard errors of 0, with a mean of $+0.037 \pm 0.016$, a split of signs as improbable as that of the tracer. The tilt term matters most in Au3, Au19 and Au27, where fitting it over $7 < R < 9.5$ kpc or $5 < R < 11.5$ kpc, with parabolas instead of straight lines, or setting it to 0 above the last slice moves their trends by at most 0.02. With the tilt term, the Jeans heights therefore grow slightly faster with age than the measured ones. Because the Jeans equation with the tilt term holds for any steady axisymmetric population, that scatter and this trend measure departures from a steady state and the noise of the measured profiles, which no prediction from these kinematics can remove. The tracer's scatter of 6.2 percent lies just above that of the Jeans heights.

3.5. *Half-mass heights*

The half-mass height needs no window, and weighs the heights above $2h_Z$ that the window leaves out. The tracer's half-mass heights lie within 10 percent of the measured ones in 57 of 70 populations, with a median ratio of 0.94 and a scatter of 8.4 percent, and exceed them in only 1 of 70. The shortfall is nearly the same at every age: the trends of the ratio over 0-10 Gyr lie within two standard errors of 0 in 6 of 7 halos, and 2.9 standard errors from it in Au19. Those over 2-10 Gyr are the filled triangles of Figure 3. With half-mass heights in place of h_Z, η is 1.15 to 1.28. The populations of every age therefore carry more of their mass far from the plane than the isothermal profile of their midplane dispersion. The Jeans heights, which take the unclipped dispersion measured at every height, account for the excess: with the measured $\sigma_z(z)$ their half-mass heights lie within 10 percent in 63 of 70 populations, and with the tilt term in 69 of 70, with a scatter of 3.7 percent. They do not separate the rise of σ_z with height from the tails of v_z that the clipping removes at the plane (Section 3.6). The sheet comes within 10 percent in 8 of 70, with a median of 1.32, so its offsets do not come from the window.

3.6. *Disk stars, finer bins and clipping*

Three checks ask whether the residual over 2-10 Gyr comes from how the populations are selected and measured. Each repeats the measurement in its own bins with 100 bootstrap resamplings. Figure 3 sets the trends of each check next to those of the tracer, so that a residual that came from the selection or the measurement would move toward 0. Keeping only the stars whose circularity $L_z/L_c(E)$ exceeds 0.7 keeps 0.07 to 1.00 of the mass of a bin, the least in the oldest bins. With these stars alone, the tracer predicts h_Z within 10 percent in 62 of 70 bins. Its trends over 0-10 Gyr lie within

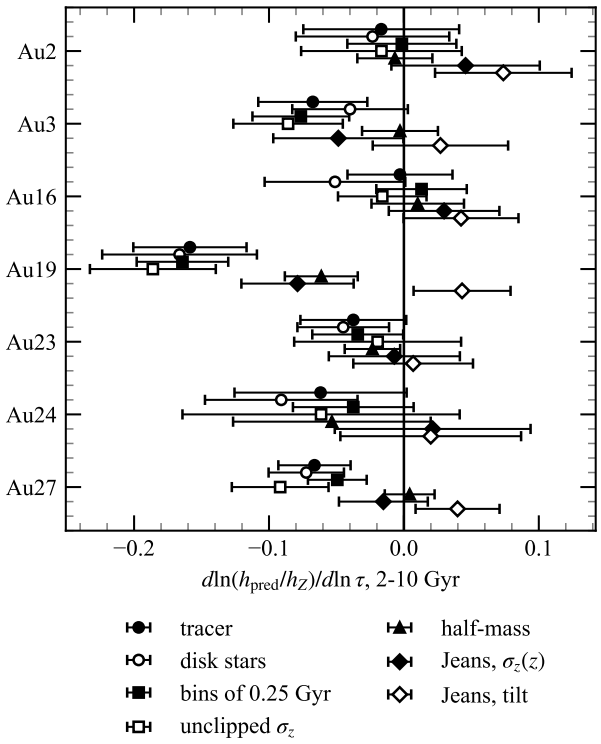


Figure 3. The ratio of the tracer to the measured heights declines slightly with age over 2-10 Gyr in every halo, a trend that the Jeans equation with the dispersions measured at every height removes. Each row is one halo, whose points give the trend $d\ln(h_{\rm pred}/h_Z)/d\ln\tau$ over 2-10 Gyr, which is 0 when a prediction follows how h_Z changes with age and negative when its ratio to h_Z declines with age. The trend of the tracer (filled circles) is negative in every halo, as it is in 7 of 7 halos with the stars on disk orbits alone (hollow circles), in 6 of 7 with bins of 0.25 Gyr (filled squares) and in 7 of 7 with the unclipped σ_z (hollow squares). With half-mass heights in place of h_Z (filled triangles), the tracer falls short by a similar amount at every age. With the measured $\sigma_z(z)$ in the Jeans equation (filled diamonds), the trends take both signs, negative in 4 of the 7 halos, with an inverse-variance mean of -0.013 ± 0.017 against -0.061 ± 0.015 for the tracer. With the tilt term as well (hollow diamonds), they are positive in 7 of the 7 halos, each within 1.5 standard errors of 0, with a mean of $+0.037 \pm 0.016$.

1.0 standard errors of 0, whereas over 2-10 Gyr 7 of 7 are negative (hollow circles). In bins of 0.25 Gyr, 6 of 7 of the trends over 2-10 Gyr are negative, up to 4.8 standard errors from 0 (filled squares). Without clipping, σ_z keeps the tails of v_z, which raise the tracer heights by 0 to 22 percent, so that some populations of 1 Gyr have heavy tails in v_z. The tracer of the unclipped dispersion comes within 10 percent in 57 of 70 bins, with trends over 2-10 Gyr that are negative in 7 of 7 halos (hollow squares). The residual is therefore a property of the disk

stars, persists in populations of 0.25 Gyr, and does not come from the clipping, which leaves the rise of σ_z with height as its cause.

3.7. *The shape of the vertical actions*

Y.-S. Ting & H.-W. Rix (2026) describe a coeval population by the distribution of its vertical actions, $p(J_z) \propto \exp[-(J_z/J_0)^{2-b}]$, which diffusion with a coefficient $\propto J_z^b$ leaves behind. Scatterers that fill the volume, such as bending waves or dark substructure, give $b = 1$, whereas scatterers confined to a thin layer at the midplane, such as giant molecular clouds, give $b = 1/2$, close to the $b = 0.51$ they measure for red-clump stars of 2-8 Gyr in the Milky Way. M. Djurić & R. Schönrich (2026) likewise derive vertical distribution functions in the actions of coeval populations scattered by giant molecular clouds, which depart from the isothermal shape. We fit b and J_0 to the vertical actions of the stars of each bin by maximum likelihood, weighted by initial mass, with the actions from the Stäckel fudge of AGAMA in the fitted potential (J. Binney 2012). An isothermal population is exponential in J_z only where the potential is harmonic: in a potential $\propto |z|^p$ its energy grows as $J_z^{2p/(p+2)}$, so a linear potential gives it $b = 4/3$. We therefore read each fitted b against $b_{\rm iso}$, the b of the isothermal tracer of the bin's σ_z in the same potential, fitted to its distribution of actions in the same way. $b_{\rm iso}$, and the heights of the population with $b = 1/2$ below, need the orbits of the population to stay inside the 12 kpc of the grid of heights on which we compute every profile, which those of 52 of 70 populations do.

In these populations the fitted b is 1.23 to 1.60 and exceeds $b_{\rm iso}$ in 52 of 52, by a median of 0.36. The cold populations, with $h_Z < 1$ kpc, have b of 1.23 to 1.57, against $b_{\rm iso}$ of 1.02 to 1.11. The vertical actions of the Auriga populations thus have heavier tails than an isothermal population of the same midplane dispersion, as their half-mass heights showed, and depart from it in the direction opposite to the thin layer. The population with $b = 1/2$ makes the same point through the heights. With J_0 set so that its stars within 0.3 kpc of the plane have the σ_z of the bin, its flat-topped velocities make it thicker than the tracer near the plane. It predicts h_Z within 10 percent in 25 of 52 populations, with a median ratio of 1.11, whereas over the same populations the tracer does so in 49 of 52, with a median of 1.02. The Auriga populations are therefore not thin-layer populations, which is not surprising in hindsight, because the star-forming gas of these simulations is a two-phase medium described by a subgrid model rather than resolved giant molecular clouds (R. J. J. Grand et al. 2017).

3.8. *The sheet*

Against the measured heights, the sheet comes within 10 percent in 8 populations, with ratios of 0.49 to 3.37: 0.51 to 0.78 for the youngest population of each halo and 2.20 to 3.37 for the oldest. In the bottom panel of Figure 1 its ratios rise with age through the band of agreement. Its trends with age over 0-10 Gyr are 0.40 to 0.82, at least 6 standard errors from 0 in every halo, as expected of a ratio that grows as $\sigma_z^{2-\eta}$, or age$^{(2-\eta)\beta_z}$ at fixed Σ. The sheet thus matches a population at intermediate ages, where $\sigma_z^2/(\pi G\Sigma)$ crosses its height, and predicts the young populations thinner and the old ones thicker than measured. This is not surprising in hindsight, because the sheet is the tracer of a layer whose own mass makes its potential, whereas a population of one age is one part of that layer and moves in the potential of all of it.

4. WHOLE DISKS

The relation of P. A. Palicio et al. (2025) uses the height of all the stars of a disk, which mixes populations of every age. If every population is an isothermal tracer, the profile of the whole disk is their sum, whose height follows from the kinematics of its populations and the potential alone. Models of the Milky Way disk are built this way, from isothermal components of different ages with dispersions prescribed by an age-velocity relation (A. C. Robin et al. 2003; A. Just & H. Jahreiß 2010; K. Sysoliatina & A. Just 2022). Here the dispersions are measured rather than prescribed, so that the sum can be tested against heights measured independently. We split the radial range of P. A. Palicio et al. (2025, Sect. 3.4) of each halo, $0.2L \le R < L$ with L from their Table 1, into rings of 1 kpc, and each ring into bins of 1 Gyr from 0 to 14 Gyr. Each cell enters as the tracer of its clipped σ_z within 0.3 kpc of the plane, in the fitted potential at the center of its ring, with the current mass of its stars, as we read the histograms of stellar mass of P. A. Palicio et al. (2025). A cell with fewer than 30 stars within 0.3 kpc of the plane has no σ_z and is left out, which removes at most 0.9 percent of the mass of a disk. Both the sum and the stars are fitted in the window of Section 2.5, with errors from 50 bootstrap draws of the stars.

One tracer per ring, with the dispersion of all its stars, predicts 0.97 to 1.13 of the measured height of the whole disk, above it in 6 of 7 disks. The sum over rings and ages predicts 0.92 to 1.04, with a median of 0.979. Figure 4 repeats the comparison in radial ranges 2 to 4 kpc wide. In the top panel, the height of the sum (dotted lines) follows the measured height of each disk (solid lines) as both rise with radius. The bottom panel

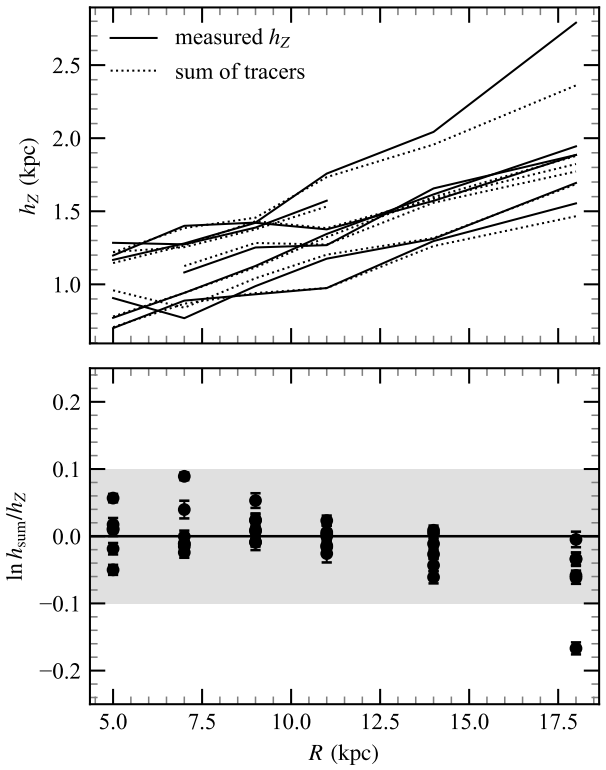


Figure 4. Summed over ages, the tracers of the populations reproduce the scale height of each whole disk and its flare with radius, within 10 percent in 38 of the 39 radial ranges. Top: the measured h_Z of all the stars of each radial range, 2 to 4 kpc wide, of the seven halos (solid lines), and the h_Z of the sum of the tracers of its populations of 1 Gyr in rings of 1 kpc (dotted lines). Bottom: the logarithm of their ratio, with errors from 50 bootstrap draws of the stars, and a gray band of agreement within 10 percent.

shows their ratio, within 10 percent in 38 of the 39 radial ranges. The heights of whole disks over the radial ranges of P. A. Palicio et al. (2025), measured in the self-consistent window rather than in their fixed windows, thus follow from the tracers of their populations.

5. THE MILKY WAY

5.1. *Star counts split by age*

A spectroscopic survey cannot give the density of a population against $|z|$ without its selection function, whereas a photometric sample can if it is complete, as we take the Gaia giants below to be. We count the Gaia DR3 (Gaia Collaboration et al. 2016, 2023) giants of a cylinder of radius 1.5 kpc around the Sun, within $|z| < 1.2$ kpc, whose reddening-free Wesenheit absolute magnitude $M_W = G - 1.9\,(G_{\rm BP} - G_{\rm RP}) + 5\log_{10}(\varpi/{\rm mas}) - 10$ lies in a window. They have parallaxes above 0.5 mas measured to better than 20 percent, $G < 16$ and $G_{\rm BP} - G_{\rm RP} > 0.9$. We take their distances as $1/\varpi$. We split each of 12 slices of $|z|$, 0.1 kpc thick, into populations with the fraction of its spectroscopic stars that belong to each, interpolated between the centers of the slices. The primary red-clump stars of LAMOST (Y. Huang et al. 2020) with ages better than 50 percent split the 461,601 Gaia stars within 0.3 mag of the red-clump peak of M_W, which we fit in each slice of $|z|$, through 22,771 matches by position and G. APOGEE DR17 (S. R. Majewski et al. 2017; Abdurro'uf et al. 2022) splits the 745,901 giants of the box $-2.6 < M_W < -0.9$ through matches by Gaia source identifier, with the ages of astroNN (J. T. Mackereth et al. 2019) (16,057 matches) and those of DistMass (A. Stone-Martinez et al. 2024), which J. Imig et al. (2025) adopt (15,933 matches). APOGEE selects its targets by magnitude and color rather than by age (G. Zasowski et al. 2013, 2017). As long as the targeting of a survey does not depend on age at a fixed position, the fraction of a population in a slice is unbiased even though the completeness of the survey changes with $|z|$, an assumption that Section 5.6 tests for both surveys. A population is a range of 2 Gyr of ages of the low-[α/Fe] disk, below [Mg/Fe] $= 0.15 - 0.10$ [Fe/H] with APOGEE and [α/Fe] $= 0.12 - 0.05$ [Fe/H] with LAMOST, and with [Fe/H] > -0.6. Its fraction varies across the cylinder as e^{gx}, with x the distance toward the Galactic center and one gradient g per population fitted to its matches.

We fit h_Z to the counts in the window of Section 2.5. The kinematics come from the giants of each population in the solar annulus $7.5 < R < 9$ kpc with distances better than 10 percent and, with APOGEE, ages better than 40 percent, with the distances and velocities of astroNN for APOGEE and of Y. Huang et al. (2020), from LAMOST radial velocities and Gaia proper motions, for LAMOST: σ_z from those with $|z| < 0.3$ kpc, clipped and with their median velocity error removed in quadrature, with every star in the Galactocentric frame of astroNN, the Sun at R_0 =8.125 kpc and 20.8 pc above the plane (M. Bennett & J. Bovy 2019). The tracer of each population, in the potential of P. J. McMillan (2017) at R_0, goes through the same counts, slice by slice, as do the sheet and the Jeans equation, so that a predicted and a measured height are fitted the same way. Errors come from 300 bootstrap draws of the Gaia stars, their matches and the kinematic stars. In the oldest bin on the DistMass scale, 30 percent of the draws give no height, because that bin lies close to the 0.6 kpc that the counts can measure.

5.2. *The tracer and the Jeans equation*

Figure 5 sets the predictions against the counts as Figure 1 does for Auriga, one panel per survey and age scale. In the potential of P. J. McMillan (2017) (hollow circles), the tracer predicts the counts with ratios of 0.85 to 0.99 with LAMOST, 0.89 to 1.14 with APOGEE on the astroNN scale and 0.92 to 1.06 on the DistMass scale, with trends over 0-10 Gyr of $+0.019 \pm 0.082$ (+0.2 standard errors), $+0.016 \pm 0.044$ (+0.4 standard errors) and -0.002 ± 0.062 (0.0 standard errors), and over 2-10 Gyr of $+0.093 \pm 0.112$ (+0.8 standard errors), -0.030 ± 0.103 (−0.3 standard errors) and -0.149 ± 0.067 (−2.2 standard errors), and η of 1.10 ± 0.17, 1.09 ± 0.11 and 1.17 ± 0.16, against 1.14 to 1.16 for the heights of the tracer over the same bins. It falls below the counts, however, in 13 of 15 bins, with medians of 0.92 to 0.94 against 1.01 in Auriga. Two causes could make it fall short: populations that are not isothermal, and a potential denser near the plane than measured. The Jeans equation with σ_z^2 measured in the slices 0-0.3, 0.3-0.6, 0.6-1.0 and 1.0-1.5 kpc, without the tilt term, which this sample cannot measure, gives heights of 0.81 to 1.12 of the counts, with medians of 0.88 to 0.91 (filled squares), 0.94 to 0.98 times the heights of the tracer in 15 of 15 bins. The dispersions of our kinematic stars rise with height in every bin, from 10-25 $\mathrm{km\,s^{-1}}$ in the slice nearest the plane to 19-28 $\mathrm{km\,s^{-1}}$ in the farthest. As in Auriga (Section 3.4), the prefactor $\sigma_z^2(0)/\sigma_z^2(z)$ of Eq. (4) lowers the density wherever the dispersion has risen, by more than the slower fall of its exponential raises it over the heights of these populations. The change of σ_z with height thus deepens the shortfall rather than removing it. The sheet, with the $77.4\,M_\odot\,\mathrm{pc^{-2}}$ of P. J. McMillan (2017) within 1.1 kpc, predicts 0.60 to 1.26 of the counts, from the youngest to the oldest bins, with trends over 0-10 Gyr of $+0.288 \pm 0.091$ (+3.2 standard errors), $+0.278 \pm 0.051$ (+5.5 standard errors) and $+0.263 \pm 0.041$ (+6.4 standard errors).

5.3. *A family of potentials*

The potential accounts for the shortfall. At R_0, the potential of P. J. McMillan (2017) has a midplane density of $0.117\,M_\odot\,\mathrm{pc^{-3}}$ and $77.4\,M_\odot\,\mathrm{pc^{-2}}$ within $|z| < 1.1$ kpc, above the $\rho_0 = 0.097 \pm 0.013\,M_\odot\,\mathrm{pc^{-3}}$ of C. F. McKee et al. (2015, their Eq. 57) and the $\Sigma_{1.1} = 68 \pm 4\,M_\odot\,\mathrm{pc^{-2}}$ of J. Bovy & H.-W. Rix (2013). Where the potential is harmonic the height of a tracer scales as $\rho_0^{-1/2}$, so a midplane density 20 percent above 0.097 thins a cold population by about 9 percent. We vary the surface density of the thin disk of P. J. McMillan (2017) by factors of 0.7, 0.8, 0.9 and 1.0, its scale

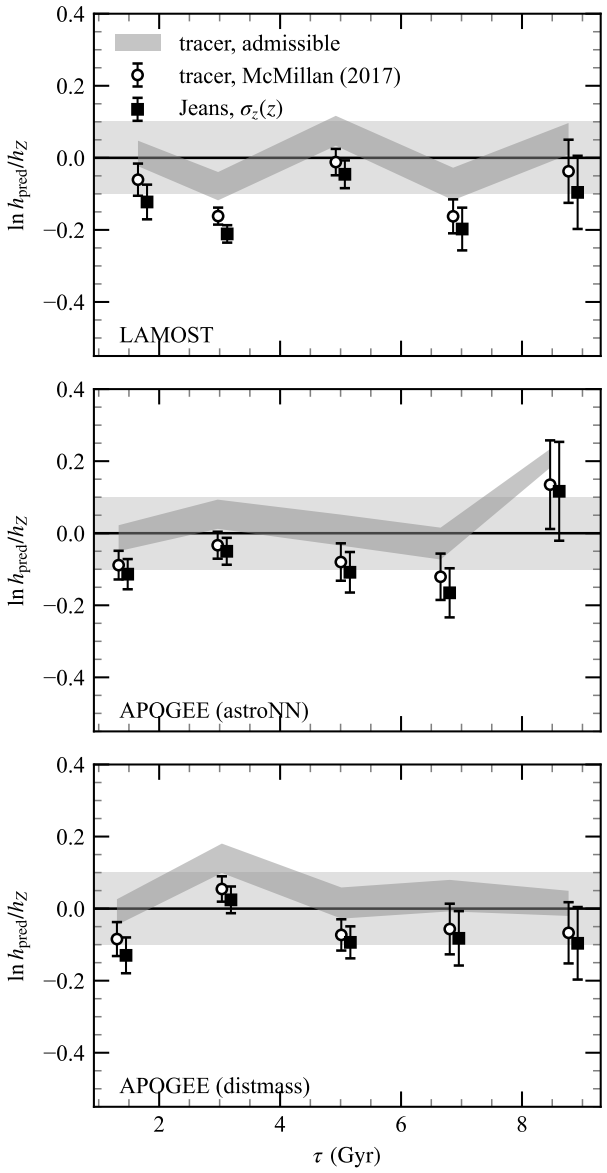


Figure 5. In the solar neighborhood the tracer follows the star counts of populations of every age with no trend with age, and reaches their level in the potentials that match the measured local density. Each panel shows, against median age, the logarithm of the predicted over the measured scale height of Gaia DR3 giants within 1.5 kpc of the Sun, split into populations of 2 Gyr by the ages of LAMOST red-clump stars (top) and of APOGEE on the astroNN (middle) and the DistMass (bottom) scales. The dark gray band spans the tracer across the 8 of 36 potentials of our family that lie within one error of the measured ρ_0 and $\Sigma_{1.1}$. In the oldest APOGEE bin on the astroNN and the DistMass scale, 2 and 1 of these potentials make the tracer thicker than the 0.6 kpc that counts below 1.2 kpc can measure, so the band there spans the others, whose tracers are the thinner ones. On the astroNN scale these give 1.20 to 1.26 times the measured height, 1.5 to 1.9 standard errors above it. Hollow circles show the tracer in the potential of P. J. McMillan (2017), whose midplane density is about 20 percent above the measured one, so that the tracer falls short, with medians of 0.92 to 0.94. Filled squares show the Jeans equation with the measured $\sigma_z(z)$, which deepens that shortfall. The light gray band marks agreement within 10 percent.

height over 0.25, 0.30 and 0.35 kpc, and the normalization of its dark halo by factors of 0.7, 1.0 and 1.3. Of these potentials, 8 of 36 lie within one error of both measurements, with ρ_0 of 0.094 to 0.108 and $\Sigma_{1.1}$ of 66.8 to 70.7. In them the tracer rises to the dark gray band of Figure 5, with medians over the counts of 0.97 to 1.07. In the oldest APOGEE bin on the astroNN and the DistMass scale, 2 and 1 of these potentials make the tracer thicker than the 0.6 kpc that the counts below 1.2 kpc can measure, so their medians leave that bin out. On the astroNN scale the other potentials predict 1.20 to 1.26 times the measured height of that bin, 1.5 to 1.9 standard errors above it if each carries the error of the tracer in the potential of P. J. McMillan (2017) there. The shortfall of the tracer can therefore come from the densities of the potential near the Sun rather than from the populations, although populations with $b = 1/2$ would match the counts in the potential of P. J. McMillan (2017) as well (Section 5.5). The family is not refitted to the rotation curve: its admissible members, whose disks or halos are lighter than those of P. J. McMillan (2017), have circular velocities at R_0 of 213 to 224 km s^{-1}, below its 233 km s^{-1}. The tracer needs only the vertical potential near the plane, which ρ_0 and $\Sigma_{1.1}$ bound.

5.4. *The local potential from the tracer*

The family also turns the test around. Every bin carries the ratio of the tracer to its h_Z in each of the 36 potentials. Across the family, the logarithm of that ratio follows a linear function of $\ln \rho_0$ and $\ln \Sigma_{1.1}$, with a root-mean-square residual over the 36 potentials of at most 1.2 percent in every bin that enters the fit below. If the populations are isothermal tracers, one potential brings every bin of a survey to its measured h_Z, which a weighted least-squares fit of the bins finds. A bin enters the fit when the tracer has a height in all 36 potentials. The oldest APOGEE bin on either age scale does not, because some of the lighter potentials make its tracer thicker than the counts can measure, so the fit takes 5, 4 and 4 bins of the three surveys. This is the inverse use of the tracer that J. Bovy & H.-W. Rix (2013) made with mono-abundance populations, with the family standing in for a parameterized potential. Figure 6 shows the midplane density that this fit returns for each survey. With the isothermal tracer (filled circles) it is 0.089 ± 0.017, 0.104 ± 0.013 and $0.120 \pm 0.030\, M_\odot\,\mathrm{pc}^{-3}$ with LAMOST and with APOGEE on the astroNN and the DistMass scale, against the $0.097 \pm 0.013\, M_\odot\,\mathrm{pc}^{-3}$ (gray band) that C. F. McKee et al. (2015) find by combining their census of stars, remnants and gas with a density of dark matter inferred from dynamical measurements of the surface density. $\Sigma_{1.1}$ is not constrained, at 104 ± 107, 69 ± 43 and $60 \pm 77\, M_\odot\,\mathrm{pc}^{-2}$, and correlates with ρ_0 at -0.92 to -0.91 in each fit: below about 1 kpc, where the populations stay, a denser midplane with less mass above it gives nearly the potential of a lighter midplane with more mass, a freedom that the errors of ρ_0 include. With LAMOST the fitted $\Sigma_{1.1}$ lies outside the 60 to 85 $M_\odot\,\mathrm{pc}^{-2}$ of the family, so that its ρ_0 extrapolates the linear dependence of the ratios on the two densities. The χ^2 per degree of freedom is 1.0 to 4.8, so with LAMOST and on the DistMass scale the bins do not all agree with one potential within their errors, as the zigzag of the LAMOST ratios with age in Figure 5 also shows. The quoted errors, scaled by the square root of the χ^2 per degree of freedom, absorb this scatter without explaining it.

5.5. *The shape of the Milky Way populations*

In the potential of P. J. McMillan (2017), at the dispersions of our bins, the isothermal tracer has b_{iso} of 1.10 to 1.17, so the $b = 0.51$ of Y.-S. Ting & H.-W. Rix (2026) departs from it by about 0.6. The population with $b = 1/2$ and the σ_z of each bin, through the same counts, gives medians of 1.03 to 1.05 in the potential of P. J. McMillan (2017) and 1.08 to 1.20 in the admissible potentials, against 0.92 to 0.94 and 0.97 to 1.07 for the isothermal tracer. Being thicker than the tracer, it exceeds the 0.6 kpc that the counts can measure in more bins: in the oldest bin of each survey it does so in 2, 8 and 8 of the admissible potentials with LAMOST and with APOGEE on the astroNN and the DistMass scale, and on the astroNN scale in the potential of P. J. McMillan (2017) as well, so these medians leave those bins out. Inverted as above, from 4, 4 and 3 bins, it needs midplane densities of 0.117 ± 0.034, 0.134 ± 0.018 and $0.146 \pm 0.071\, M_\odot\,\mathrm{pc}^{-3}$ (hollow circles in Figure 6). Being thicker than the isothermal population of the same midplane dispersion, it fits the counts best in potentials denser than the measurements near the Sun, close to that of P. J. McMillan (2017), by 0.6, 1.6 and 0.7 combined standard errors. At the measured densities the counts favor the isothermal shape, although the error of ρ_0 keeps the preference weak, so the star counts of the solar neighborhood do not by themselves decide the shape that Y.-S. Ting & H.-W. Rix (2026) measure in the actions.

5.6. *Age errors and selection*

The LAMOST ages, which we keep up to a relative error of 50 percent, carry typical errors of 30 percent (Y. Huang et al. 2020). The APOGEE ages are cut at a relative error of 40 percent for the kinematic stars

only, so the matches that split the counts keep every age. Bins of 2 Gyr therefore mix their neighbors. The targeting of either survey could also depend on age at a fixed position. Two checks remeasure every survey, with 100 bootstrap draws per bin. The first draws every age again from its error, lognormal with the relative error as the dispersion of ln age, which doubles the variance of the errors, in 4 realizations. The medians of the tracer over the counts become 0.89 to 0.95, against 0.92 to 0.94 with the ages as given. Its trends over 0-10 Gyr become −0.006 to 0.089, within 1.9 standard errors of 0. Doubling the variance of the age errors thus leaves the test where it was, presumably because the mixing of ages enters the counts and the kinematics of a bin alike.

The second asks whether the fraction of a population at a fixed position depends on the magnitude of its stars, as it could if the targeting depended on age there. In each slice of $|z|$ we split the matches of each survey at their median G, and measure each survey with each half. The heights of the bright over the faint half are 0.81 to 1.29, within two standard errors in 13 of 15 bins and at most 2.7 standard errors apart: 5 of 5 with LAMOST, and 8 of 10 with APOGEE on either age scale. A targeting of APOGEE that depends on age at a fixed position may thus move some of its bins by up to the largest of these differences, whereas the LAMOST fractions show no dependence on magnitude. The test detects a targeting by age only where it also changes with magnitude.

6. COMPARISON WITH IMIG ET AL.

J. Imig et al. (2025) fit every population of APOGEE DR17 with more than 100 stars, in bins of [α/M], [M/H] and DistMass age, with a model of the whole footprint of the survey corrected for its selection function. Their model of a population is $\nu(R,z) = \nu_\odot \Sigma(R) \exp[-|z|/h_Z(R)]$, with a broken exponential $\Sigma(R)$ and a scale height that flares linearly. Their heights at the Sun are therefore the scales of exponentials fitted to the whole range of $|z|$ that APOGEE reaches, whereas ours are windowed heights of counts below $|z| = 1.2$ kpc. An isothermal population is flat at the plane and falls more steeply above, so the exponential that matches its mean height is thinner than its windowed height: for the tracer of P. J. McMillan (2017) at R_0, the windowed height is 1.21 to 1.51 times the exponential fit below 1.2 kpc for σ_z of 10 to 25 km s^{-1}. We therefore compare each of their populations with our counts of the same cell of age and [M/H], both fitted with the same exponential below $|z| = 1.2$ kpc, their model put through our cylinder and counts. We take their populations below the [α/M] of their Eq. (1), give each cell the σ_z of its giants within 0.3 kpc of the plane,

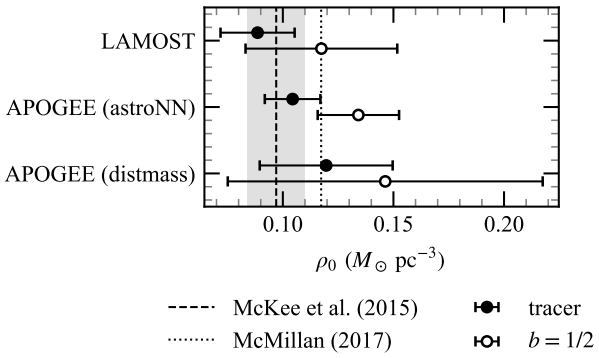


Figure 6. The star counts measure the local midplane density once the potential is fitted to them rather than taken from a model. If every population is an isothermal tracer, the density that fits the counts of each survey agrees with the measurement of C. F. McKee et al. (2015). Points show the ρ_0 at which the bins of a survey match their counts, with one-sigma errors, from the 5, 4 and 4 bins of the three surveys that have a predicted height in every potential of our family (4, 4 and 3 for $b = 1/2$), for the isothermal tracer (filled) and for populations whose vertical actions are distributed as $\exp[-(J_z/J_0)^{3/2}]$, the $b = 1/2$ of Y.-S. Ting & H.-W. Rix (2026) (hollow), which fit best in a potential denser than measured, near that of P. J. McMillan (2017) (dotted line), although their errors keep this preference weak. The gray band and the dashed line mark the $0.097 \pm 0.013\, M_\odot$ pc^{-3} of C. F. McKee et al. (2015).

and measure a growth at fixed metallicity as the slope of $\ln h_Z$ against $\ln \sigma_z$ with one intercept per bin of [M/H]. Of their cells, 40 of 101 have at least 100 matched stars and a height in both our counts and their model.

Fitted the same way, our counts over their model have a median of 1.05 over 40 cells, and change by $+0.033 \pm 0.085$ (+0.4 standard errors) per dex of metallicity at fixed age, so the two measurements agree in level and in how the heights depend on metallicity. The two differ in how the heights grow. At fixed metallicity our exponential fits over theirs change by $+0.206 \pm 0.029$ (+7.1 standard errors) per ln age, from a median of 0.88 over 6 cells below 2.51 Gyr to one of 1.19 over 16 cells above 5.01 Gyr. Figure 7 shows how the heights grow with the σ_z of the giants of each cell. Our exponential heights (filled circles) grow as $\sigma_z^{1.26\pm0.09}$ at fixed metallicity, whereas their published heights (hollow circles) grow as $\sigma_z^{0.92\pm0.08}$, consistent with $h_Z \propto \sigma_z$ (dotted line), the growth of an isothermal population in a harmonic potential. The exponential fit to the tracer, however, grows with exponents of 1.47 to 1.52 between 10 and 25 km s^{-1} across the admissible potentials of Section 5.3 (dashed lines), and of 1.30 to 1.34 over the whole profile, because the hotter populations reach heights where the potential is no longer harmonic,

and the cut at 1.2 kpc steepens the growth further. Our exponential heights also grow more slowly than this fit to the tracer, by 2.4 to 3.0 standard errors, whereas the windowed heights of our age bins grow as the tracer does (Section 5.2). We have not traced the difference between the two estimates. A potential that is denser or lighter near the plane within the measured errors moves the heights of all the populations by similar factors, which changes their level more than their growth, so the uncertainty of the potential does not account for the difference. The targeting of APOGEE does not account for the growth of our heights either, because split by LAMOST, whose targeting and ages differ, our exponential fits over theirs change by $+0.183 \pm 0.019$ ($+9.5$ standard errors) per ln age. This tests our side of the comparison only, because their heights rest on the selection function of APOGEE.

Part of the difference can come from the range of the fits. Their fit takes the stars at every height that APOGEE reaches, whereas ours stops at 1.2 kpc. For the tracer, the exponential fit to the whole profile is 0.87 to 1.00 times the fit below 1.2 kpc, the lowest for the hottest population, so their old heights should come out thinner than ours by up to that factor. That would account for part of the excess of our old cells, but not for the young cells, where our fits are the thinner. We have not traced the rest. Both measurements use the same survey on the same age scale, so the errors of the ages mix the same neighbors into both, whereas the two measurements differ in where the stars lie: their height at the Sun comes from a model of the whole footprint, in which each population has one exponential in $|z|$ at every radius and a linear flare, whereas ours comes from the stars within 1.5 kpc of the Sun. A fit of their model to the stars near the Sun alone would show whether the difference comes from the rest of the footprint.

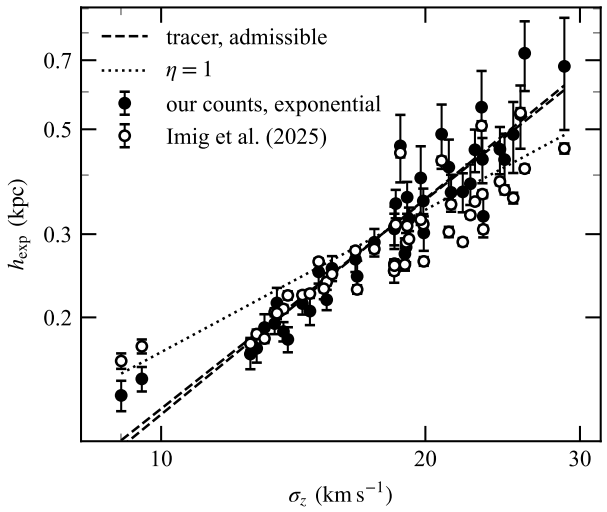


Figure 7. Our star counts and the heights that J. Imig et al. (2025) published for the solar neighborhood agree in level but differ in how they grow with the dispersion, theirs more slowly than ours, which in turn grow more slowly than the exponential fit to the tracer in the potentials of our family that match the measured local density. Our exponential fits below $|z| = 1.2$ kpc to the counts of each cell of age and [M/H] of J. Imig et al. (2025) (filled circles) and their published heights at the Sun (hollow circles) are plotted against the σ_z of the giants of the cell. Ours grow as $\sigma_z^{1.26\pm0.09}$ at fixed metallicity, theirs as $\sigma_z^{0.92\pm0.08}$. The dashed lines show the growth of the exponential fit to the tracer in those potentials, with exponents of 1.47 to 1.52, and the dotted line $h_Z \propto \sigma_z$, the growth in a harmonic potential, all drawn through the mean of our fits.

7. DISCUSSION

7.1. *Age-thickness relations*

The heights of mono-age populations grow with age as age$^{\eta\beta_z}$, rather than as the age$^{2\beta_z}$ of the sheet. Over 0-10 Gyr in the Auriga halos, σ_z grows with exponents β_z of 0.48 to 0.94 and h_Z with exponents β_h of 0.55 to 1.11. The products $\eta\beta_z$ are 0.56 to 1.05, whereas $2\beta_z$ would be 0.96 to 1.88. In the Milky Way our counts give β_z of 0.42 to 0.48 and β_h of 0.46 to 0.53, against $\eta\beta_z$ of 0.45 to 0.53 and $2\beta_z$ of 0.83 to 0.96. Because η and β_h come from the same heights, they would agree exactly if the heights and the dispersions followed power laws in age, so their agreement is a consistency of the fits rather than a test. $\beta_h - \eta\beta_z$ is -0.01 to 0.13, largest in Au19 and Au27, whose oldest populations thicken abruptly. The test is that η follows the local slope of the tracer of each potential (Section 3.3). The exponent of the sheet exceeds that of the tracer by a factor of $2/\eta$, about 1.6-1.8 here. M. Martig et al. (2026, Sect. 5.3) note that the sheet would make the half-mass heights of the stars of lenticular galaxies grow with age with an exponent between 0.8 and 1, unlike the heights they measure. A tracer lowers that exponent by the factor $2/\eta$, although no value of η produces the flat half-mass heights of their younger populations, so the tracer does not by itself account for their measurements.

7.2. *The shape of coeval populations*

Two departures from isothermality bear on this work, in opposite directions. In Auriga, the dispersion of every population grows with height (Section 3.4), the populations carry more mass far from the plane than the isothermal profile of their midplane dispersion (Section 3.5), and their vertical actions have heavier tails than those of an isothermal population (Section 3.7). In the Milky Way, Y.-S. Ting & H.-W. Rix (2026) find

actions with a sharper cutoff, $b = 0.51$, which they attribute to heating by molecular clouds. S. Sharma et al. (2021) find that the dispersions of stars of fixed age and metallicity rise with height, whereas J. Bovy et al. (2012a) find mono-abundance populations nearly isothermal in $|z|$. The dispersions of our Milky Way kinematic stars rise with height in every bin (Section 5.2). Our star counts alone do not separate these shapes: the population with $b = 1/2$ fits them best in a potential denser at the midplane than measured, close to that of P. J. McMillan (2017), whereas the isothermal tracer fits them at the measured density. For the heights, the shape is a correction of order 10 percent at fixed σ_z, small against the ratios of 0.49 to 3.37 between the sheet and the measured heights, whereas for the dynamics of disk heating the shape is the record that Y.-S. Ting & H.-W. Rix (2026) read.

7.3. *The radial action and the scale height*

For a population of one age, the slope $a = J_{R,\mathrm{med}}/h_Z$ of the relation between radial action and scale height grows with age as $\mathrm{age}^{2\beta_R - \eta\beta_z}$, with $\sigma_R \propto \mathrm{age}^{\beta_R}$, for a tracer in a fixed potential, rather than the $\mathrm{age}^{2(\beta_R - \beta_z)}$ of the sheet at fixed Σ, so the derivation of the slope through the sheet, made for whole disks, does not carry over to single populations. Across galaxies, the heights of whole disks follow from the tracers of their populations (Section 4). With a proxy for the reference radial action $J_{R,w}$ of P. A. Palicio et al. (2025), the median of the cells of a face-on map of the median J_R over $0.2L \leq R < L$, with L the radial extent of each galaxy in their Table 1, the slope against the self-consistent heights of the seven halos is 72 ± 28 km/s. Their fit across 23 disks gives 3.69×10^{-2} in units of the angular momentum of the Sun per kpc, or $72.6\,\mathrm{km\,s^{-1}}$ with the $8.249\,\mathrm{kpc} \times 238.5\,\mathrm{km\,s^{-1}}$ that P. A. Palicio et al. (2023) take for it. It depends, however, on one halo, Au2, without which it is 25 ± 15 km/s, so seven halos do not test the relation across galaxies.

7.4. *Limitations*

The Auriga snapshots are at the standard resolution of the public release, whose stellar softening of 369 pc (R. J. J. Grand et al. 2017) is comparable to the heights of 0.40 to 0.62 kpc of the populations younger than 1 Gyr. R. Pakmor et al. (2025) find that the rise of the velocity dispersion at large radii seen at that resolution might be entirely numerical. Numerical heating raises the dispersion and the height of a population together, so it shifts both sides of the test, but it could change η and the shape of the actions, which the Level 3 runs of the release, at higher resolution, would test. Every Auriga population comes from one annulus per halo, so we do not know how η depends on radius. The fitted potential is axisymmetric, which the potential of the snapshot averaged over azimuth checks but cannot make exact for the bar of Au23. In the Milky Way, the family of potentials varies the thin disk and the halo of P. J. McMillan (2017) but neither its thick disk nor its gas, and is not refitted to the rotation curve. The surface density of J. Bovy & H.-W. Rix (2013) comes from stars modeled in equilibrium, so the agreement of the tracer with the counts remains one of consistency with that assumption. The inversion constrains ρ_0 but not $\Sigma_{1.1}$. The Jeans equation of the Milky Way leaves out the tilt term, which held up the old populations of Au19. The fractions of LAMOST come from primary red-clump stars, whereas the Gaia stars within 0.3 mag of the red-clump peak also include giants of other stages, whose mix of ages can differ and change with $|z|$. Finally, we have not found why our heights and those of J. Imig et al. (2025) grow differently with age.

8. CONCLUSIONS

1. In seven Auriga disks, the isothermal tracer in a potential fitted to all the particles predicts the scale heights of 64 of 70 populations of 1 Gyr within 10 percent, with a scatter of 6.2 percent, just above the 5.5 percent of the vertical Jeans equation with the measured dispersion profile and tilt term. The self-gravitating sheet does so for 8, predicting young populations thinner and old ones thicker than measured.

2. The heights grow as σ_z^η with η of 1.10 to 1.26, which equals within two standard errors the local slope of the tracer of the potential in 6 of 7 halos, so age-thickness relations grow as $\mathrm{age}^{\eta\beta_z}$ rather than $\mathrm{age}^{2\beta_z}$.

3. The populations are not exactly isothermal. Their dispersions rise with height, they carry more mass far from the plane than the isothermal profile of their midplane dispersion, and their vertical actions have heavier tails. The ratio of the tracer to their heights declines slightly with age over 2-10 Gyr, with the disk stars alone and in bins of 0.25 Gyr as well, which the Jeans equation with the measured $\sigma_z(z)$ removes, except in Au19, where it needs the tilt term. With the tilt term, the Jeans heights grow slightly faster with age than the measured ones.

4. Summed over ages, the tracers predict 0.92 to 1.04 of the heights of whole disks and follow their flare.

5. In the solar neighborhood, Gaia DR3 star counts split by LAMOST and APOGEE ages follow the tracer with no trend with age in the potential of P. J. McMillan (2017), at medians of 0.97 to 1.07 in potentials that match the measured local densities, and imply midplane densities of 0.089 ± 0.017, 0.104 ± 0.013 and $0.120 \pm 0.030\, M_\odot\, \mathrm{pc}^{-3}$. Doubling the variance of the age errors leaves the ratios of the tracer to the counts in the potential of P. J. McMillan (2017), and their trends with age, in place. Splitting the age tracers at their median magnitude moves no LAMOST height by more than two standard errors, and 8 of 10 APOGEE heights by less than two, the others by up to 2.7.

6. Fitted the same way, our counts and the model of J. Imig et al. (2025) agree in level and in the dependence of the heights on metallicity, but differ in how the heights grow with age, a difference that neither the potential nor the targeting of APOGEE behind our counts accounts for and that the range of the fits accounts for only in part.

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We use APOGEE DR17, part of the Sloan Digital Sky Survey IV (M. R. Blanton et al. 2017; S. R. Majewski et al. 2017; Abdurro'uf et al. 2022; J. C. Wilson et al. 2019; J. E. Gunn et al. 2006), with the astroNN (H. W. Leung & J. Bovy 2019a,b; J. T. Mackereth et al. 2019) and DistMass (A. Stone-Martinez et al. 2024) value-added catalogs. Funding for the Sloan Digital Sky Survey IV has been provided by the Alfred P. Sloan Foundation, the U.S. Department of Energy Office of Science, and the Participating Institutions. SDSS-IV acknowledges support and resources from the Center for High Performance Computing at the University of Utah. The SDSS website is www.sdss4.org. SDSS-IV is managed by the Astrophysical Research Consortium for the Participating Institutions of the SDSS Collaboration including the Brazilian Participation Group, the Carnegie Institution for Science, Carnegie Mellon University, Center for Astrophysics | Harvard & Smithsonian, the Chilean Participation Group, the French Participation Group, Instituto de Astrofísica de Canarias, The Johns Hopkins University, Kavli Institute for the Physics and Mathematics of the Universe (IPMU) / University of Tokyo, the Korean Participation Group, Lawrence Berkeley National Laboratory, Leibniz Institut für Astrophysik Potsdam (AIP), Max-Planck-Institut für Astronomie (MPIA Heidelberg), Max-Planck-Institut für Astrophysik (MPA Garching), Max-Planck-Institut für Extraterrestrische Physik (MPE), National Astronomical Observatories of China, New Mexico State University, New York University, University of Notre Dame, Observatório Nacional / MCTI, The Ohio State University, Pennsylvania State University, Shanghai Astronomical Observatory, United Kingdom Participation Group, Universidad Nacional Autónoma de México, University of Arizona, University of Colorado Boulder, University of Oxford, University of Portsmouth, University of Utah, University of Virginia, University of Washington, University of Wisconsin, Vanderbilt University, and Yale University.

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We use the LAMOST-Gaia red-clump catalog of Y. Huang et al. (2020). Guoshoujing Telescope (the Large Sky Area Multi-Object Fiber Spectroscopic Telescope LAMOST) is a National Major Scientific Project built by the Chinese Academy of Sciences. Funding for the project has been provided by the National Development and Reform Commission. LAMOST is operated and managed by the National Astronomical Observatories, Chinese Academy of Sciences.

We use the density fits of J. Imig et al. (2025), released under CC BY 4.0 at https://github.com/astrojimig/mw_density_imig2025.

APPENDIX

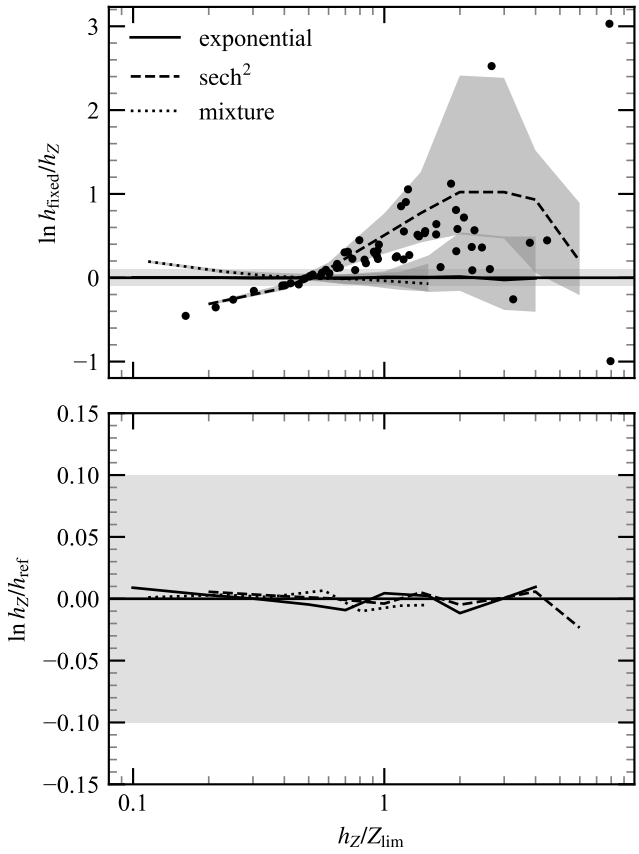


Figure 8. A window of fixed size biases the scale height of any profile that is not exponential, by an amount that depends on how thick the profile is against the window, whereas the self-consistent window $L = 2h_Z$ recovers every injected profile within 2.4 percent. Top: the logarithm of the height fitted in a fixed window $Z_{\rm lim} = 1\,$kpc over the self-consistent height, against $h_Z/Z_{\rm lim}$, for exponentials (solid), sech^2 layers (dashed) and mixtures of two exponentials (dotted), each the median of 40 draws of 5,000 heights with a dark gray band between the 16th and 84th percentiles, and for the Auriga populations (dots), each fitted in the fixed window of its halo, with $Z_{\rm lim}$ of 0.5 to 2.5 kpc, which follow the sech^2 curve where they are thin against the window. Bottom: the logarithm of the self-consistent height of the injected draws over that of the exact, noise-free profile. The light gray bands mark agreement within 10 percent.

A. THE WINDOW OF THE FIT

The scale height of a profile that is not exponential depends on the window in which it is fitted. A sech$^2(z/z_0)$ layer is flat for $|z| \ll z_0$ and falls as $\exp(-2|z|/z_0)$ above, so a window much wider than z_0 returns a scale close to the mean height of the whole profile, $z_0 \ln 2$, whereas a window narrower than z_0 holds an almost flat profile, which the fit reads as a very large scale, or as no scale at all when the mean height reaches $L/2$. We drew 40 samples of 5,000 heights from an exponential, a sech^2 layer, and a mixture of two exponentials, and fitted each in a fixed window, $L = Z_{\rm lim} = 1\,$kpc, and in the self-consistent window $L = 2h_Z$. Figure 8 shows the ratio of the two heights against the thickness of the profile relative to the window (top), and the self-consistent height against that of the exact, noise-free profile (bottom). The fixed window recovers exponentials within 2.7 percent of their scale, as the maximum-likelihood estimate of an exponential should, whereas it returns 0.73 times the self-consistent height of a sech^2 layer that is thin against the window and 1.66 times that of one as thick as the window. For profiles thicker than the window, the fixed window finds no height in up to 58 percent of the draws, which the medians of Figure 8 leave out. The self-consistent window recovers every injected profile within 2.4 percent. The Auriga populations (dots in the top panel), each fitted in the fixed window of its halo, follow the sech^2 curve: below $h_Z = 0.5\,Z_{\rm lim}$ their median ratio of the fixed to the self-consistent height is 0.91, against 0.89 for sech^2 and 1.00 for an exponential, as expected of tracers in a potential

that is harmonic near the plane. The windows of P. A. Palicio et al. (2025, Table 1), with $Z_{\rm lim}$ of 0.5 to 2.5 kpc chosen for each galaxy, are fixed in this sense.

The self-consistent window takes the smallest L that satisfies $L = 2h(L)$. With few stars inside $L/2$, the profile of a hot population is nearly flat inside a small window, so that the noise of the fitted $1/h$ can satisfy the condition by chance, far below the true scale. In 3000 draws of 300 stars from an exponential of $h = 3$ kpc, a minimum of 20 stars inside $L/2$ gave a root below 1 kpc in 25 draws, a minimum of 50 in 1 and a minimum of 100 in none, which is the minimum we use.

B. THE MOCK GALAXY AND THE POTENTIAL

We tested the analysis on a mock snapshot of a known axisymmetric galaxy, written in the units and the file layout of Auriga, tilted, moving, and close to two faces of the periodic box. The analysis recovers the spin axis within 0.012 degrees, the circular velocity of the annulus within 0.3 percent, $\Delta\Phi$ at the median radius within 1.0 percent over $0.05 < z \leq 1.5$ kpc, and the height and the median radial action of every bin within 0.3 percent and 0.7 percent of the truth. In the real snapshots, the potential averaged over eight sectors of azimuth differs from the axisymmetric fit by -68 to 44 $(\mathrm{km\,s^{-1}})^2$ between the plane and $z = 1$ kpc, and by at most 300 $(\mathrm{km\,s^{-1}})^2$ up to 5.7 kpc, the last slice we measure, above which we hold the difference fixed. The tracer in the potential of the snapshot is therefore 0.97 to 1.04 times the tracer in the fit, even with the bar of Au23.

REFERENCES

Abdurro'uf, Accetta, K., Aerts, C., et al. 2022, ApJS, 259, 35, doi: 10.3847/1538-4365/ac4414

Bahcall, J. N. 1984, ApJ, 276, 169, doi: 10.1086/161601

Benítez-Llambay, A., Navarro, J. F., Frenk, C. S., & Ludlow, A. D. 2018, MNRAS, 473, 1019, doi: 10.1093/mnras/stx2420

Bennett, M., & Bovy, J. 2019, MNRAS, 482, 1417, doi: 10.1093/mnras/sty2813

Bienaymé, O., Robin, A. C., & Crézé, M. 1987, A&A, 180, 94

Binney, J. 2012, MNRAS, 426, 1324, doi: 10.1111/j.1365-2966.2012.21757.x

Blanton, M. R., Bershady, M. A., Abolfathi, B., et al. 2017, AJ, 154, 28, doi: 10.3847/1538-3881/aa7567

Bovy, J., & Rix, H.-W. 2013, ApJ, 779, 115, doi: 10.1088/0004-637x/779/2/115

Bovy, J., Rix, H.-W., Hogg, D. W., et al. 2012a, ApJ, 755, 115, doi: 10.1088/0004-637x/755/2/115

Bovy, J., Rix, H.-W., Liu, C., et al. 2012b, ApJ, 753, 148, doi: 10.1088/0004-637x/753/2/148

Bovy, J., Rix, H.-W., Schlafly, E. F., et al. 2016, ApJ, 823, 30, doi: 10.3847/0004-637x/823/1/30

Deemer, Jr., W. L., & Votaw, Jr., D. F. 1955, Ann. Math. Stat., 26, 498, doi: 10.1214/aoms/1177728494

Djurić, M., & Schönrich, R. 2026, MNRAS, 552, stag1628, doi: 10.1093/mnras/stag1628

Gaia Collaboration, Prusti, T., de Bruijne, J. H. J., et al. 2016, A&A, 595, A1, doi: 10.1051/0004-6361/201629272

Gaia Collaboration, Vallenari, A., Brown, A. G. A., et al. 2023, A&A, 674, A1, doi: 10.1051/0004-6361/202243940

Garbari, S., Read, J. I., & Lake, G. 2011, MNRAS, 416, 2318, doi: 10.1111/j.1365-2966.2011.19206.x

Grand, R. J. J., Fragkoudi, F., Gómez, F. A., et al. 2024, MNRAS, 532, 1814, doi: 10.1093/mnras/stae1598

Grand, R. J. J., Springel, V., Gómez, F. A., et al. 2016, MNRAS, 459, 199, doi: 10.1093/mnras/stw601

Grand, R. J. J., Gómez, F. A., Marinacci, F., et al. 2017, MNRAS, 467, 179, doi: 10.1093/mnras/stx071

Gunn, J. E., Siegmund, W. A., Mannery, E. J., et al. 2006, AJ, 131, 2332, doi: 10.1086/500975

Holmberg, J., & Flynn, C. 2000, MNRAS, 313, 209, doi: 10.1046/j.1365-8711.2000.02905.x

Holmberg, J., Nordström, B., & Andersen, J. 2009, A&A, 501, 941, doi: 10.1051/0004-6361/200811191

Huang, Y., Schönrich, R., Zhang, H., et al. 2020, ApJS, 249, 29, doi: 10.3847/1538-4365/ab994f

Imig, J., Holtzman, J. A., Zasowski, G., Lian, J., et al. 2025, ApJ, 990, 203, doi: 10.3847/1538-4357/adf723

Jia, Y., Chen, Y., Du, C., & Zhao, G. 2023, A&A, 669, A107, doi: 10.1051/0004-6361/202142977

Jia, Y., Yang, C., Chen, Y., Du, C., & Zhao, G. 2024, A&A, 692, A167, doi: 10.1051/0004-6361/202452294

Just, A., & Jahreiß, H. 2010, MNRAS, 402, 461, doi: 10.1111/j.1365-2966.2009.15893.x

Kapteyn, J. C. 1922, ApJ, 55, 302, doi: 10.1086/142670

Kuijken, K., & Gilmore, G. 1989a, MNRAS, 239, 571, doi: 10.1093/mnras/239.2.571

Kuijken, K., & Gilmore, G. 1989b, MNRAS, 239, 605, doi: 10.1093/mnras/239.2.605

Leung, H. W., & Bovy, J. 2019a, MNRAS, 483, 3255, doi: 10.1093/mnras/sty3217

Leung, H. W., & Bovy, J. 2019b, MNRAS, 489, 2079, doi: 10.1093/mnras/stz2245
Ludlow, A. D., Fall, S. M., Schaye, J., & Obreschkow, D. 2021, MNRAS, 508, 5114, doi: 10.1093/mnras/stab2770
Mackereth, J. T., Bovy, J., Leung, H. W., et al. 2019, MNRAS, 489, 176, doi: 10.1093/mnras/stz1521
Mackereth, J. T., Bovy, J., Schiavon, R. P., et al. 2017, MNRAS, 471, 3057, doi: 10.1093/mnras/stx1774
Majewski, S. R., Schiavon, R. P., Frinchaboy, P. M., et al. 2017, AJ, 154, 94, doi: 10.3847/1538-3881/aa784d
Martig, M., Minchev, I., & Flynn, C. 2014, MNRAS, 443, 2452, doi: 10.1093/mnras/stu1322
Martig, M., Pinna, F., Falcón-Barroso, J., et al. 2026, MNRAS, 547, stag458, doi: 10.1093/mnras/stag458
McKee, C. F., Parravano, A., & Hollenbach, D. J. 2015, ApJ, 814, 13, doi: 10.1088/0004-637x/814/1/13
McMillan, P. J. 2017, MNRAS, 465, 76, doi: 10.1093/mnras/stw2759
Mirabal, D., Gallart, C., Ruiz-Lara, T., et al. 2026, ApJL, 1010, L38, doi: 10.3847/2041-8213/aeb0a3
Oort, J. H. 1932, BAN, 6, 249
Oort, J. H. 1960, BAN, 15, 45
Pakmor, R., Fragkoudi, F., Grand, R. J. J., et al. 2025, MNRAS, 543, 4355, doi: 10.1093/mnras/staf1755
Palicio, P. A., Recio-Blanco, A., Poggio, E., et al. 2023, A&A, 670, L7, doi: 10.1051/0004-6361/202245026
Palicio, P. A., Recio-Blanco, A., Tepper-García, T., et al. 2025, A&A, 695, A193, doi: 10.1051/0004-6361/202453611
Robin, A. C., Reylé, C., Derrière, S., & Picaud, S. 2003, A&A, 409, 523, doi: 10.1051/0004-6361:20031117
Roškar, R., Debattista, V. P., & Loebman, S. R. 2013, MNRAS, 433, 976, doi: 10.1093/mnras/stt788
Sharma, S., Hayden, M. R., Bland-Hawthorn, J., et al. 2021, MNRAS, 506, 1761, doi: 10.1093/mnras/stab1086
Sivertsson, S., Read, J. I., Silverwood, H., et al. 2022, MNRAS, 511, 1977, doi: 10.1093/mnras/stac094
Skrutskie, M. F., Cutri, R. M., Stiening, R., et al. 2006, AJ, 131, 1163, doi: 10.1086/498708
Spitzer, Jr., L. 1942, ApJ, 95, 329, doi: 10.1086/144407
Stone-Martinez, A., Holtzman, J. A., Imig, J., et al. 2024, AJ, 167, 73, doi: 10.3847/1538-3881/ad12a6
Sysoliatina, K., & Just, A. 2021, A&A, 647, A39, doi: 10.1051/0004-6361/202038840
Sysoliatina, K., & Just, A. 2022, A&A, 666, A130, doi: 10.1051/0004-6361/202243780
Ting, Y.-S., & Rix, H.-W. 2019, ApJ, 878, 21, doi: 10.3847/1538-4357/ab1ea5
Ting, Y.-S., & Rix, H.-W. 2026, arXiv e-prints, arXiv:2608.12156
van der Kruit, P. C. 1988, A&A, 192, 117
van der Kruit, P. C., & Searle, L. 1981, A&A, 95, 105
Vasiliev, E. 2019, MNRAS, 482, 1525, doi: 10.1093/mnras/sty2672
Wilson, J. C., Hearty, F. R., Skrutskie, M. F., et al. 2019, PASP, 131, 055001, doi: 10.1088/1538-3873/ab0075
Zasowski, G., Cohen, R. E., Chojnowski, S. D., et al. 2017, AJ, 154, 198, doi: 10.3847/1538-3881/aa8df9
Zasowski, G., Johnson, J. A., Frinchaboy, P. M., et al. 2013, AJ, 146, 81, doi: 10.1088/0004-6256/146/4/81